

Mirage

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Figure 1: Mirage, Video Installation

Abstract

Mirage is a computational film installation that explores how generative AI can fictionalize temporal sequences within global surveillance systems. Constructed from real surveillance frames captured via public webcam infrastructure, *Mirage* interpolates the time gaps between frames using AI-generated visual and narrative content. Rather than creating fictional worlds from scratch, the work embeds speculative micro-narratives into the neutral flow of surveillance imagery—raising critical questions about visual truth and the narrative potential of machinic vision.

CCS Concepts

• Applied computing → Media arts;

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Keywords

AI Art, Generative AI, LLMs, Narrative, Cinema

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1 Background

In recent years, the rise of AI-generated imagery has challenged conventional understandings of visual authenticity. From deepfakes [7] to modern text-to-image models, AI has introduced a form of photorealism that disrupts the image's historical role as evidence. As Chesney and Citron argue, synthetic media are reshaping public trust in visual proof and journalistic integrity [2]. Baudrillard's concept of simulacra is especially relevant here: when images circulate as closed systems of signs, detached from any verifiable referent, the boundary between real and fabricated dissolves [1]. Generative AI, trained on vast datasets, assembles statistically coherent visuals rather than representing specific events. The question shifts from truthfulness to the algorithmic production of visual credibility.

Surveillance imagery, by contrast, has long been associated with factual documentation. As a functional medium, it exists not to express but to monitor, archive, and discipline. Foucault's panopticism frames surveillance as power through visibility [6], while Farocki's operative images highlight its task-oriented nature [5]. These perspectives frame surveillance as objective and non-narrative. *Mirage* subverts this role by transforming surveillance into material for fiction. AI-generated narratives emerge from real surveillance frames, producing speculative scenes through inference, genre cues, and stylistic synthesis—blurring fiction and data-driven logic.

Through theatrical intervention, *Mirage* questions how AI reshapes the construction of truth. By embedding fiction into ostensibly factual footage, it reveals the erosion between record and construct. As Zylinska notes, when image-making shifts from human to algorithmic agency, the image becomes computational rather than perceptual [8]. In a post-truth context—where affective narratives often override facts—*Mirage* stages a visual allegory: images no longer reflect the world, but script it. Here, AI acts not just as a tool of synthesis, but as a meaning-making system embedded in infrastructures of surveillance and control.



Figure 2: A screenshot of a story imagined by an AI based on surveillance from Miami Beach.

2 Technical Description

Mirage utilizes video frames extracted from eight publicly accessible surveillance cameras located across different global cities. These cameras broadcast live imagery through publicly accessible online platforms. The current version of the work spans eight continuous hours, with each hour corresponding to a distinct geographic location (e.g., Venice, London, Sapporo). In each hour, six independent narrative segments—each structured as a ten-minute episodic unit—are generated. While each segment presents a unique storyline, all six share a consistent spatial setting derived from the original surveillance scene. This structure enables *Mirage* to investigate the interplay between spatial repetition and narrative variation, challenging the monotony of functional surveillance footage.

For each ten-minute sequence, the system preserves the initial and final frames as authentic surveillance stills. The intervening eight minutes are generated through an AI pipeline that produces both visual content and character-driven dialogue. GPT-4o

is prompted as a screenwriter, tasked with generating a coherent narrative based on the two anchor frames and a predefined cinematic genres. The output is segmented into scene-level descriptions, which are then translated into synthetic images using AI-based image generation techniques. In parallel, voiceovers are synthesized using ElevenLabs [4], incorporating cultural tone, regional accent, and emotional nuance.

The generated material is rendered and played through a custom pipeline built in TouchDesigner [3]. Each segment is revealed via a progressive line-scan technique, simulating the gradual construction of visual information by a machine. By simulating the visual characteristics of machine-based processing, this approach highlights the algorithmic foundations of image construction and underscores the operational logic of machine vision in AI-generated imagery. The resulting experience resembles that of watching an AI-mediated live television broadcast—where viewers are witnessing a sequence of speculative events being dynamically narrated by the system.

3 Artist Statement

Mirage explores how AI reshapes visual truth by transforming real-time surveillance footage into speculative fiction. Drawing on theories of simulation, algorithmic vision, and post-truth media, the work uses generative models to script and visualize narratives within systems originally designed for observation, not storytelling. By embedding fictional content into these functional frameworks, *Mirage* highlights the fading boundary between documentation and construction. It invites reflection on how machine vision participates in the authorship of reality.

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